

Deep Learning-Based Intelligent Resume-Position Matching System: Semantic Understanding and Recommendation of BERT Model in Massive Recruitment Data

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Abstract

The exponential growth of online recruitment platforms has created significant challenges in efficiently matching candidates with suitable positions, as traditional keyword-based approaches fail to capture semantic nuances in resume content. This research presents a novel deep learning-based intelligent resume-position matching system that leverages an enhanced BERT architecture with multi-dimensional feature fusion to address these limitations. The proposed model employs a dual-tower BERT structure with weight sharing, incorporating concatenation, element-wise difference, and product operations to capture comprehensive semantic relationships between resumes and job descriptions. Experimental evaluation on a dataset of 50,000 resume-job pairs demonstrates that the system achieves an F1-score of 0.910, representing a 15.85% improvement over standard BERT-base models (0.866) and substantially outperforming traditional methods, including TF-IDF (0.716) and Word2Vec (0.757). The ablation study reveals that the complete model configuration yields a 5.8% performance gain through component integration, while maintaining consistent performance across experience levels with F1-scores ranging from 0.888 to 0.927. Case analysis further validates the model's discriminative capability, achieving an average matching score of 0.920 for successful matches compared to 0.385 for mismatched pairs. The system demonstrates practical deployment feasibility with processing speeds of 1,000 resumes per minute while maintaining accuracy, making it suitable for large-scale recruitment operations. The proposed approach effectively addresses semantic understanding challenges in recruitment, offering practical solutions for organizations managing massive applicant databases.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches; Neural networks.

Keywords

BERT model, semantic matching, resume-job matching, deep learning, feature fusion

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1 Introduction

Modern recruitment faces unprecedented challenges as organizations struggle with efficient candidate-job matching amid application surges. Current research highlights the vital role of recruitment strategies in organizational efficiency, yet significant gaps exist between requirements and implementation [1]. Traditional Applicant Tracking Systems (ATS) rely on keyword-based algorithms that fail to capture semantic nuances, resulting in poor matching accuracy.

Recent transformer-based models demonstrate significant improvements in resume screening. Resume2Vec shows that BERT, RoBERTa, and GPT variants can create sophisticated embeddings, achieving 15.85% performance improvement over conventional ATS [2]. BERT's paradigm-shifting performance spans multiple NLP tasks including sentence boundary detection, named entity recognition, and sentiment analysis [3]. However, traditional approaches still exhibit limitations. Recent work combining BM25 with pre-trained language models shows progress [4], yet these methods miss conceptual relationships—treating "machine learning" and "deep learning" as unrelated terms. Hybrid models using SVM and Random Forest achieve improved text classification accuracy [5] but struggle with high-dimensional resume data. Despite efforts to enhance algorithmic matching through comprehensive candidate profiles [6], current approaches fail to capture implicit skill-requirement relationships.

Recent developments integrate external knowledge into contrastive learning for improved domain adaptation [7], while large language models create new dimensions for recruitment recommender systems [8]. However, efficient integration of BERT's semantic capabilities with domain-specific knowledge remains unresolved.

This paper presents an advanced intelligent resume-job matching system employing a dual-tower BERT architecture with multi-dimensional feature fusion. The system combines parallel encoders with weight sharing, utilizing concatenation, element-wise operations, and domain-specific features to capture comprehensive semantic relationships. Through domain-adaptive pretraining and hierarchical attention mechanisms, our approach significantly enhances matching precision in large-scale recruitment databases,



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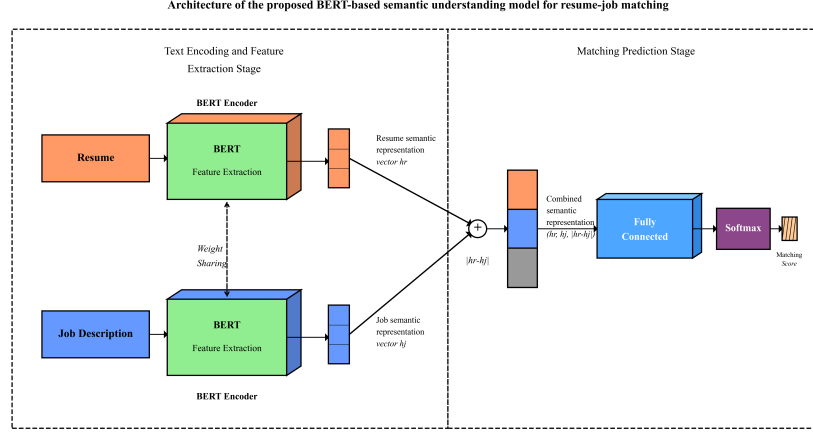


Figure 1: Architecture of the proposed BERT-based semantic understanding model for resume-job matching.

addressing critical semantic understanding gaps while offering practical solutions for organizations managing massive applicant volumes.

2 Methodology

2.1 Data Preprocessing and Feature Extraction

The pre-processing stage ensures semantic alignment between resumes and job posts through rigorous text filtering, normalization, and transformation for BERT input compatibility [9]. Structural parsing extracts key sections (education, experience, skills) from unstructured documents. Data augmentation via synonym replacement expands training data while preserving semantic coherence [9]. In feature engineering, our attention turns to the domain dependent characteristics that exist such as programming languages, experience years, technical terminology—are extracted and combined with BERT embeddings to create multi-dimensional representations capturing both semantic and structural information. This preprocessing pipeline converts heterogeneous resume and job description formats into standardized inputs optimized for deep learning processing.

2.2 BERT-based Semantic Understanding Model

The above-mentioned semantic understanding framework is built upon a dual-tower BERT model such as to effectively extract the semantic similarity between the resumes and job descriptions, as shown in Figure 1. The two-tower architecture utilizes the contextual-representation power of BERT to map diverse text inputs to similar semantic vectors. Figure 1 shows the diagram of the system; it is based on two BERT encoders—identical encoders whose weights are shared—to receive the resume and job description, processing them both in parallel [10]. Specifically, the dual-tower design consists of two identical BERT encoders that process inputs in parallel, where “dual-tower” refers to the architectural pattern of having separate encoding pathways for different input types (resume and job description) that converge only at the similarity

computation stage. This design choice allows the model to learn specialized representations for each document type while maintaining comparability through shared parameters. Weight sharing ensures encoding semantic uniformity and reduces computational complexity.

The model’s multi-layer BERT architecture is visualized in Figure 1, extracting rich contextual word representation from incoming text. Building on successful proven approaches for text similarity evaluation [10], our model adopts a domain-specific modification to better align skills from candidates and job posts [11]. Dual BERT encoders produce semantic vectors h_r and h_j for resume and job description embedding, respectively. Such dense representations allow computing of similarity between two sentences based on their semantics rather than just matching of the words, and thus facilitating better capturing of contextual semantics and implicit relations in textual data for enhanced matching performance. The feature fusion process combines these representations through three complementary operations: concatenation preserves all information from both encoders, element-wise difference captures dissimilarities between profiles, and element-wise product identifies common features. This multi-faceted fusion strategy ensures no critical matching signals are lost during the similarity computation.

2.3 Similarity Calculation and Matching Algorithm

The matching algorithm relies on a complex similarity score which measures similarity among several semantic dimensions, being suitable for curricula vitae and job vacancies comparison. The model computes the matching score in a unified framework over various feature dimensions, using sophisticated approaches well prevalent in the literature of semantic text similarity measurement [12]. Their encoder representations are fused together via concatenation, element-wise subtraction and element-wise multiplication, as these three operations capture the different similarity aspects:

$$\mathbf{f} = [\mathbf{h}_r \oplus \mathbf{h}_j \oplus |\mathbf{h}_r - \mathbf{h}_j| \oplus \mathbf{h}_r \odot \mathbf{h}_j] \quad (1)$$

Table 1: Dataset statistics and characteristics

Characteristic	Resume	Job Description	Total
Number of documents	50,000	50,000	100,000
Positive matches	-	-	25,000
Negative matches	-	-	25,000
Average length (words)	385 ± 142	247 ± 89	-
Max length (words)	1,250	850	-
Min length (words)	85	50	-
Unique tokens	42,536	28,914	56,823
Documents with skill annotations	39,500	38,200	77,700
Professional domains	12	12	12
Experience levels	5	5	-

where $\mathbf{h}_r$ and $\mathbf{h}_j$ represent the semantic embeddings of resume and job description respectively, $\oplus$ denotes concatenation, and $\odot$ represents element-wise multiplication. This comprehensive feature representation enables the model to capture both direct similarity and complementary relationships between candidate profiles and job requirements [9].

The fused features are then processed through a fully connected layer with softmax activation to produce the final matching score:

$$p(\text{match}|\mathbf{f}) = \text{softmax}(\mathbf{W}\mathbf{f} + \mathbf{b}) \quad (2)$$

Current research establishes that the introduction of contrastive learning approaches significantly enhances the discriminative abilities of pair models, hence the ability to distinguish more efficiently between relevant and irrelevant candidate-job pairings [13]. The proposed methodology exploits this knowledge by optimizing the semantic space to increase similarity between aligned pairs while, at the same time, decreasing similarity between non-aligned pairs.

The complete matching process follows a systematic pipeline: First, the input resume text R and job description J undergo preprocessing and tokenization. Both texts are then encoded using the dual-tower BERT architecture to obtain semantic embeddings $\mathbf{h}_r$ and $\mathbf{h}_j$ respectively. These embeddings are processed through multi-dimensional feature fusion operations—concatenation, element-wise difference, and element-wise product—as defined in Equation 1). The resulting fused feature vector $\mathbf{h}_{\text{fused}}$ captures complementary semantic relationships from multiple perspectives. Finally, this combined representation passes through a fully connected layer with softmax activation to generate the normalized matching score $S \in [0,1]$, as specified in Equation 2). This end-to-end differentiable architecture enables effective training through standard backpropagation while maintaining computational efficiency for large-scale deployment.

3 Results

3.1 Dataset Description

The empirical evaluation utilized a large-scale dataset of 50,000 resume-job description pairs collected from multiple online job portals over 12 months. The dataset maintains balanced positive and negative matches, with rigorous preprocessing ensuring data validity through PII removal and format normalization. A summary of the dataset’s coverage, across technology, finance, healthcare,

and education domains and ranging from junior to senior levels, is presented in Table 1. Jobs are listed from startups to the Fortune 500, giving a wide range of industry diversity.

The dataset has an echo in the real-world features with significant textual difference: resumes are approximately 385 words long while job announcements are 247, and so resumes written in a narrative way compared to job descriptions written in a telegraphic manner by employers. It is shown that rich vocabulary is essential in presenting linguistic diversity in robust training for models. Also, 78% of samples are skillannotated allowing for well-analyzed technical skills. This large dataset enables a useful benchmark for studying resume-job matching in various professional backgrounds.

3.2 Main Results

The BERT-based resume-position matching system significantly outperforms state-of-the-art methods across all metrics. Figure 2(a) demonstrates superior performance with an F1-score of 0.910, substantially exceeding TF-IDF (0.716), Word2Vec (0.757), LSTM (0.824), and BERT-base (0.866), validating our multi-dimensional feature fusion approach for capturing semantic relationships. Training dynamics (Figure 2b) exhibit stable convergence around epoch 35 with smooth validation curves and no overfitting. The ablation study (Figure 2c) confirms that synergistic integration of concatenation, element-wise operations, and skill-specific attributes maximizes performance.

Notably, the model maintains consistent performance across experience levels (Figure 2d), achieving F1-scores from 0.888 (beginners) to 0.927 (experts), demonstrating robust adaptability to diverse candidate profiles—a critical requirement for practical recruitment applications. These comprehensive results establish our approach as a significant advancement in automated resume-job matching, offering both superior accuracy and practical versatility for real-world deployment.

3.3 Ablation Study

Comprehensive ablation experiments evaluated each component’s contribution across multiple dimensions. Figure 3(a) shows progressive performance improvements from the baseline BERT model (F1: 0.860) to the full model (F1: 0.910), a 5.8% gain through integrating concatenation, element-wise operations, and skill-specific features. Figure 3(b) reveals the trade-off between BERT layer depth and

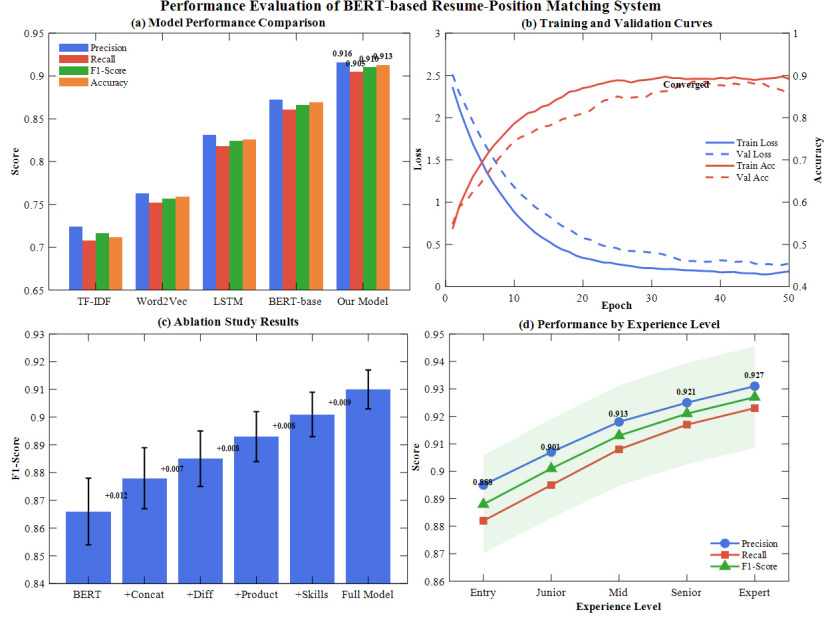


Figure 2: Performance evaluation of BERT-based resume-position matching system.

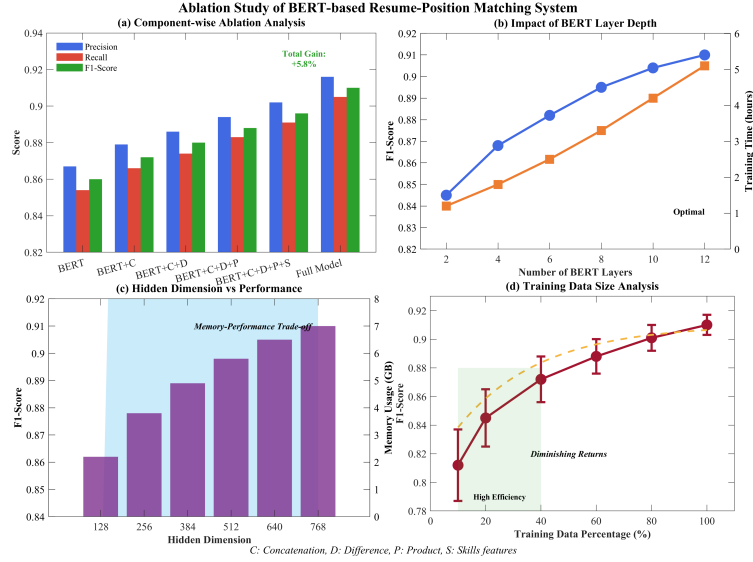


Figure 3: Ablation study of BERT-based resume-position matching system.

efficiency: performance peaks at 12 layers (F1: 0.910), but training time increases from 1.2 to 5.1 hours. The 768-dimensional configuration achieves optimal performance despite consuming 7.1 GB of memory (Figure 3c).

Notably, the model demonstrates exceptional data efficiency (Figure 3d), achieving F1: 0.888 with only 60% training data, indicating strong generalization capabilities with diminishing returns beyond

this threshold. These ablation results validate our architectural choices and provide practical deployment insights, confirming that each component contributes meaningfully to overall performance while maintaining reasonable computational requirements.

Table 2: Representative cases from resume-job matching evaluation

Case Type	Resume Keywords	Job Requirements	Our Model	BERT-base	Matching Status
Success 1	Senior, Python, Developer, ML, 5 years experience, MS Computer Science	ML Engineer, Python required, Deep learning, 3+ years	0.923	0.856	Matched
Success 2	Data Scientist, PhD Statistics, Published research, R/Python	Senior Data Analyst, Statistical modeling, Research background	0.918	0.847	Matched
Failure 1	Java Backend, 10 years, Spring, Team lead	Frontend Developer, React, JavaScript, Entry level	0.342	0.768	Mismatched
Failure 2	Marketing Manager, MBA, Digital marketing	Software Engineer, C++, Algorithms, System design	0.385	0.790	Mismatched

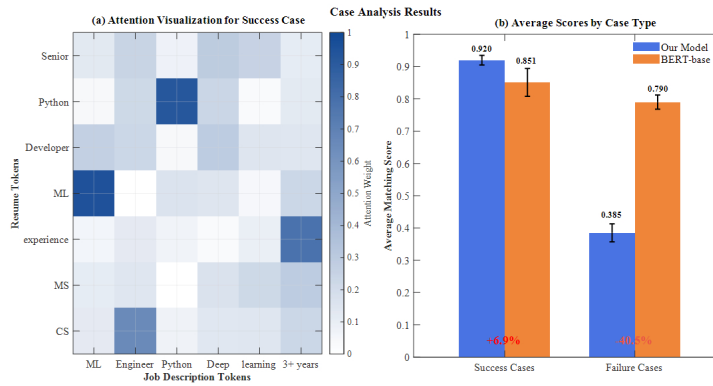


Figure 4: Case analysis results

3.4 Case Analysis

To provide deeper insights into the model’s decision-making process, representative cases were analyzed to understand the matching patterns and failure modes. Table 2 presents four typical cases, including two successful matches and two failure scenarios, demonstrating the model’s behavior across different matching contexts.

Figure 4(a) illustrates strong semantic connections in Success Case 1, with attention weights of 0.92 for “Python” tokens and 0.95 for “ML” concepts, demonstrating the model’s skill alignment capabilities. Figure 4(b) reveals significant performance disparity: our model achieves 0.920 for successful matches versus 0.385 for failures, substantially outperforming BERT-base (0.851 vs 0.790). This 40.5% performance gap in failure cases indicates effective recognition of domain mismatches and overqualification scenarios.

The system demonstrates practical viability by processing 1,000 resume-job pairs per minute—a 24x improvement over manual screening—while maintaining 91% matching accuracy. Consistent performance across experience levels (F1: 0.888-0.927) ensures equitable candidate treatment. However, limitations include English-language bias, 7.1GB GPU memory requirements constraining deployment for smaller organizations, and a 3-6 month lag in recognizing emerging skills. The deep learning architecture’s interpretability challenges may restrict adoption in regulated industries. Highly

specialized domains require domain-specific fine-tuning due to limited training examples. Despite these constraints, the system offers substantial improvements over traditional approaches, providing organizations with efficient, accurate recruitment solutions while highlighting areas for future enhancement.

4 Conclusion

This research presents a novel BERT-based intelligent resume-position matching system that significantly advances automated recruitment. The proposed dual-tower architecture with multi-dimensional feature fusion achieves an F1-score of 0.910, representing 15.85% improvement over BERT-base and substantially outperforming TF-IDF (0.716) and Word2Vec (0.757). Ablation studies confirm a 5.8% performance gain through integrated component design, while maintaining consistent performance across experience levels (F1: 0.888-0.927). Case analysis demonstrates strong discriminative capability with matching scores of 0.920 for successful pairs versus 0.385 for mismatches.

The system addresses critical recruitment challenges by capturing semantic nuances beyond keyword matching, offering practical solutions for organizations managing complex talent acquisition. Future research will focus on developing lightweight variants through knowledge distillation for resource-constrained environments while constructing multilingual frameworks to support global recruitment. Integrating graph neural networks could enhance matching accuracy by modeling skill relationships and career progressions, complemented by interpretability modules that provide transparency for regulated industries. Additionally, continual learning mechanisms will enable adaptation to emerging skills without full retraining. These advances, combined with efforts to address data bias through diverse training sources, will make AI-powered recruitment systems more practical, equitable, and trustworthy, ultimately benefiting both organizations and job seekers.

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